LIFTING SMOOTH MORPHISMS ETALE-LOCALLY THROUGH A THICKENING

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1098-1111 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 1113-1208, the complete author proof (94 body lines).

Editorial note. Navigation stubs: 13 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 5 of queue_stacks-stacks-more-morphisms.json (pb=4226, stmt 1098-1111, proof 1113-1208).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 063J: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1208–1208. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 063G: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1208–1208. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 062I: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1208–1208. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CER: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1208–1208. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0CEM: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1208–1208. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 062H: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1208–1208. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 062F: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1208–1208. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 069G: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1208–1208. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04II: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1208–1208. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03FX: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1208–1208. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CFV: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1208–1208. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06U9: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1208–1208. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CJ6: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1208–1208. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{X} \subset \mathcal{X}'$ be a thickening of algebraic stacks. Let W be an algebraic space and let $W \to \mathcal{X}$ be a smooth morphism. There exists an étale covering $\{W_i \to W\}_{i \in I}$ and for each i a cartesian diagram*

$$\begin{array}{ccc} W_i & \longrightarrow & W_i' \\ \downarrow & & \downarrow \\ \mathcal{X} & \longrightarrow & \mathcal{X}' \end{array}$$

with $W_i' \to \mathcal{X}'$ smooth.

Proof. Choose a scheme U' and a surjective smooth morphism $U' \to \mathcal{X}'$. As usual we set $U = \mathcal{X} \times_{\mathcal{X}'} U'$. Then $U \to \mathcal{X}$ is a surjective smooth morphism. Therefore the base change

$$V = W \times_{\mathcal{X}} U \longrightarrow W$$

is a surjective smooth morphism of algebraic spaces. By Topologies on Spaces, Lemma 0CFV we can find an étale covering $\{W_i \to W\}$ such that $W_i \to W$ factors through $V \to W$. After covering W_i by affines (Properties of Spaces, Lemma 03FX) we may assume each W_i is affine. We may and do replace W by W_i which reduces us to the situation discussed in the next paragraph.

Assume W is affine and the given morphism $W \to \mathcal{X}$ factors through U. Picture

$$W \xrightarrow{i} U \to \mathcal{X}$$

Since W and U are smooth over $\mathcal{X}$ we see that i is locally of finite type (Morphisms of Stacks, Lemma 06U9). After replacing U by $\mathbf{A}^n_U$ we may assume that i is an immersion, see Morphisms, Lemma 04II. By Morphisms of Stacks, Lemma 0CJ6 the morphism i is a local complete intersection. Hence i is a Koszul-regular immersion (as defined in Divisors, Definition 063J) by More on Morphisms, Lemma 069G.

We may still replace W by an affine open covering. For every point $w \in W$ we can choose an affine open $U'_w \subset U'$ such that if $U_w \subset U$ is the corresponding affine open, then $w \in i^{-1}(U_w)$ and $i^{-1}(U_w) \to U_w$ is a closed immersion cut out by a Koszul-regular sequence $f_1, \ldots, f_r \in \Gamma(U_w, \mathcal{O}_{U_w})$. This follows from the definition of Koszul-regular immersions and Divisors, Lemma 063G. Set $W_w = i^{-1}(U_w)$; this is an affine open neighbourhood of $w \in W$. Choose lifts $f'_1, \ldots, f'_r \in \Gamma(U'_w, \mathcal{O}_{U'_w})$ of $f_1, \ldots, f_r$. This is possible as $U_w \to U'_w$ is a closed immersion of affine schemes. Let $W'_w \subset U'_w$ be the closed subscheme cut out by $f'_1, \ldots, f'_r$. We claim that $W'_w \to \mathcal{X}'$ is smooth. The claim finishes the proof as $W_w = \mathcal{X} \times_{\mathcal{X}'} W'_w$ by construction.

To check the claim it suffices to check that the base change $W'_w \times_{\mathcal{X}'} X' \to X'$ is smooth for every affine scheme X' smooth over $\mathcal{X}'$. Choose an étale morphism

$$Y' \to U'_w \times_{\mathcal{X}'} X'$$

with Y' affine. Because $U'_w \times_{\mathcal{X}'} X'$ is covered by the images of such morphisms, it is enough to show that the closed subscheme Z' of Y' cut out by $f'_1, \ldots, f'_r$ is smooth over X'. Picture

$$\begin{array}{ccccc}
Z' & \longrightarrow & Y' & & \\
\downarrow & & \downarrow & & \\
W'_w \times_{\mathcal{X}'} X' & \longrightarrow & U'_w \times_{\mathcal{X}'} X' & \longrightarrow & X' \\
\downarrow & & \downarrow & & \\
W'_w = V(f'_1, \ldots, f'_r) & \longrightarrow & U'_w & &
\end{array}$$

Set $X = \mathcal{X} \times_{\mathcal{X}'} X'$, $Y = X \times_{X'} Y' = \mathcal{X} \times_{\mathcal{X}'} Y'$, and $Z = Y \times_{Y'} Z' = X \times_{X'} Z' = \mathcal{X} \times_{\mathcal{X}'} Z'$. Then $(Z \subset Z') \to (Y \subset Y') \subset (X \subset X')$ are (cartesian) morphisms of thickenings of affine schemes and we are given that $Z \to X$ and $Y' \to X'$ are smooth. Finally, the sequence of functions $f'_1, \ldots, f'_r$ map to a Koszul-regular sequence in $\Gamma(Y', \mathcal{O}_{Y'})$ by More on Algebra, Lemma 062H because $Y' \to U'_w$ is smooth and hence flat. By More on Algebra, Lemma 0CER (and the fact that Koszul-regular sequences are quasi-regular sequences by More on Algebra, Lemmas 062F, 0CEM, and 062I) we conclude that $Z' \to X'$ is smooth as desired. □